A Complete Small-Signal Equivalent Circuit Model of Cooled Butterfly-Type 2.5 Gbps DFB Laser Modules and its Application to Improve High Frequency Characteristics

Jaehoon Lee, Seungki Nam, S. H. Lee, and Jichai Jeong, *Senior Member, IEEE*

***Abstract*—A small-signal equivalent circuit model of 2.5 Gbps DFB laser modules with butterfly-type dual-in-line packages has been proposed and verified using extracted parameters. Parameters related to the equivalent circuit have been extracted from measured S parameters using the modified two-port black box model. This model includes small-signal equivalent circuits of components used for 2.5 Gbps DFB laser modules such as DFB laser, coplanar waveguides, matching resistor, bonding wires, and thermoelectric cooler (TEC). From this equivalent circuit modeling, we show that calculated frequency characteristics of DFB lasers on submount and complete DFB laser modules are similar to their measured frequency characteristics, respectively. Based on this equivalent circuit model, we propose and demonstrate a method that can improve frequency characteristics of 2.5 Gbps DFB laser modules through both experiments and simulations.**

***Index Terms*—Distributed feedback lasers, equivalent circuits, frequency response, simulation.**

I. Introduction

LASER diode modules with the direct modulation technique have been widely used as a key component for optical transmission systems for their simplicity [1]. Optical transmission performance using the direct modulation is highly dependent on high frequency characteristics of laser diode modules, because they determine optical pulse shapes at the output of high data rate transmitters [2], [3]. Frequency characteristics of laser diode modules are dependent on package parasitic components as well as characteristics of semiconductor lasers. In addition, the interaction between the semiconductor laser and parasitic components due to electrical packages and TEC affects frequency characteristics of laser diode modules. To understand frequency characteristics of laser diode modules and then improve them, it is needed to accurately model equivalent circuit representing the whole characteristics of DFB laser modules including not only characteristics of semiconductor lasers and parasitic components of packages but also their interactions.

In this paper, we model and verify a complete small signal equivalent circuit of cooled butterfly 2.5 Gbps DFB laser modules including characteristics of DFB lasers, parasitic components related to electrical packages and TEC, and their interactions. Parameters related to the equivalent circuit are extracted from measured S parameters. Based on the equivalent circuit model, we propose and demonstrate a method to improve high frequency characteristics of 2.5 Gbps DFB laser modules. Section II includes the method to extract parameters related to the small-signal equivalent circuit of 2.5 Gbps DFB laser modules from measured S parameters. In Section III, improvement of frequency characteristics is described using the equivalent circuit model with additional connection of a chip capacitor parallel to the thin-film resistor. Finally, Section IV includes conclusions of our investigations.

Manuscript received March 1, 2002; revised November 1, 2002.

J. Lee, S. Nam, and J. Jeong are with the Department of Radio Engineering, Korea University, Seoul 136-701, Korea.

S. H. Lee is with the Optoelectronics Division, Samsung Electronics, Suwon 442-742, Korea.

Digital Object Identifier 10.1109/TADVP.2002.807562

II. Parameter Extraction of Small-Signal Equivalent Circuit of Cooled Butterfly-Type 2.5 Gbps DFB Laser Modules

A. Description of 2.5 Gbps DFB Laser Modules

A DFB laser module, shown in Fig. 1, consists of a DFB laser, tapered coplanar waveguide on submount, coplanar waveguide for the interconnection of microwave signals from pin 12 of butterfly packages to DFB lasers, TEC under the submount, optical assembly for coupling light into fibers, and bonding wires for electrical connections. TEC is located at the bottom of the submount. Coplanar waveguides were used to reduce the impedance mismatching. Several bonding wires were used to connect the microwave feed-line from DFB lasers to pin 12 of butterfly packages via tapered coplanar and coplanar waveguides. Coplanar waveguide includes a 20 Ω thin-film resistor to become the total impedance of around 25 Ω including impedance of DFB lasers (typical impedance: 5 Ω) for 25 Ω impedance matching. The ground of this waveguide was connected to the case ground of the butterfly-type package through a 100 pF chip capacitor.

B. Parameter Extraction Method of Equivalent Circuit From Measured S Parameters

To extract equivalent circuit parameters of 2.5 Gbps DFB laser modules from measured S parameters, we used the optimization procedure of a commercial microwave simulator, Serenade. A modified two-port black box model shown in Fig. 2 was used to model an equivalent circuit of 2.5 Gbps DFB laser modules. We modified a conventional two-port black box model

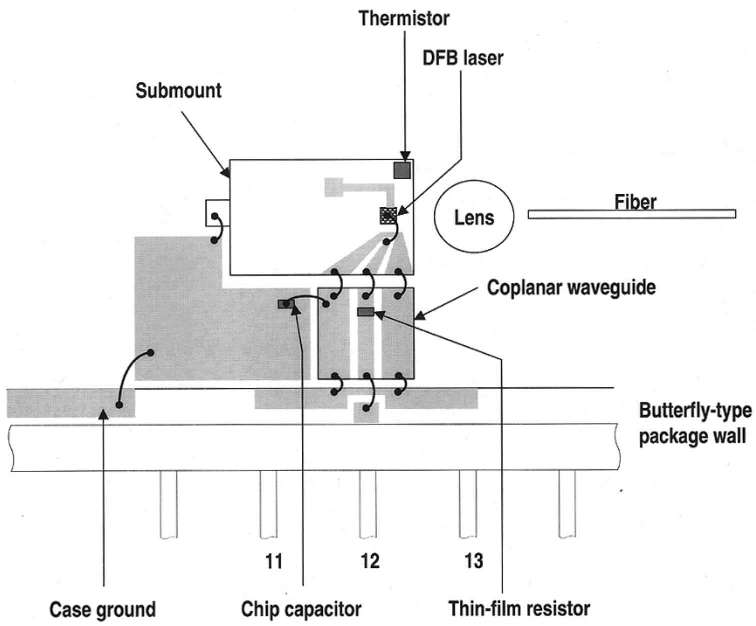


Fig. 1. Schematic of 2.5 Gbps DFB laser modules.

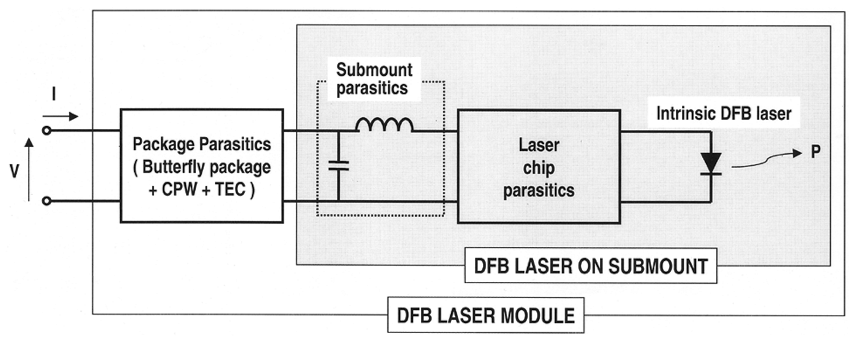


Fig. 2. Two-port black box model of 2.5 Gbps DFB laser modules.

[4], [5] to be compatible with our case. The two-port black box model of the equivalent circuit of DFB laser modules can be divided into two boxes; one box associated with DFB laser on submount and the other box associated with coplanar waveguide, TEC, and butterfly package of DFB laser modules. The box model associated with DFB laser on submount includes equivalent circuits of intrinsic DFB laser, parasitic components of laser chip, and parasitic components of submount.

Before putting a DFB laser on submount into butterfly packages, S parameters were measured using an Agilent 8703A lightwave component analyzer and a GSG-type microwave probe. For measurements of S parameters of DFB lasers on submount, electrical and optical ports were calibrated. Electrical port 1 using a GSG-type microwave probe was calibrated by open, short, and load calibration standards on the standard calibration substrate. Electrical through on the standard calibration substrate and optical through were measured for the complete calibration. For measurements of S parameters of cooled butterfly 2.5 Gbps DFB laser modules, the standard calibration kits to calibrate electrical ports were used, instead of using the standard calibration substrate. For measurements of S parameters, the GSG-type microwave probe was located at the end of the tapered coplanar waveguide. From this measurement, S parameters included only frequency characteristics of DFB laser, tapered coplanar waveguide, bonding wires, and other parasitic components of submount. After assembling a complete DFB laser module using the DFB laser on submount, we measured S parameters for DFB laser modules through pin 11 (ground), 12 (signal), and 13 (ground) with the proper DC bias condition.

C. Equivalent Circuit of DFB Laser on Submount Including Tapered Coplanar Waveguide

Fig. 3 shows the equivalent circuit model of DFB laser on submount. The small-signal ac equivalent circuit was used to model the intrinsic laser [6], because we measured frequency responses, S_{21}, of DFB laser on submount. This model can be obtained by linearizing the rate equation with ac and dc components. The capacitance C_t is a sum of the space charge capacitance and the active layer diffusion capacitance. The inductance L_s represents the small signal photon storage. Other circuit elements, R_1, R_{s1}, and R_{s2} are related to dc components in the rate equation. The parasitic components due to the semiconductor

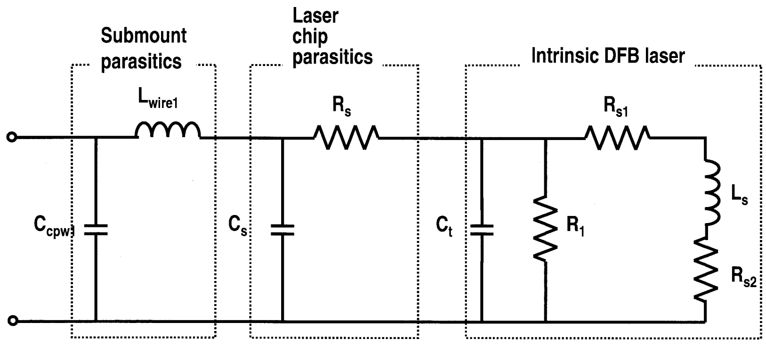


Fig. 3. Small-signal equivalent circuit model of DFB laser on the submount.

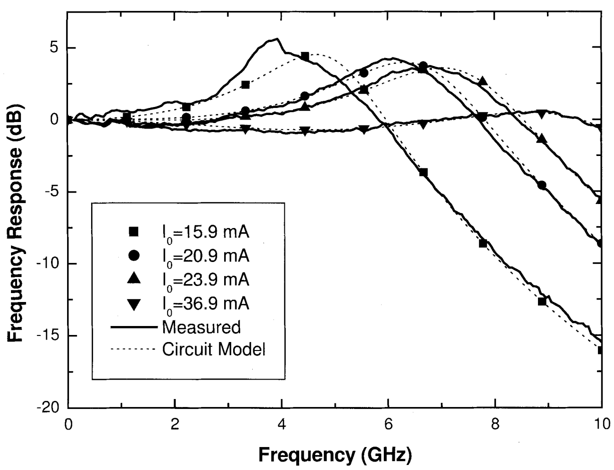


Fig. 4. Frequency responses of DFB laser on submount with different DC bias currents: Solid lines represent measured data and dotted lines represent simulated data.

TABLE I
EXTRACTED PARAMETERS OF THE EQUIVALENT CIRCUIT FOR DFB LASER ON SUBMOUNT EXCEPT FOR INTRINSIC DFB LASER SHOWN IN FIG. 3

C_{cpw1}	0.145 pF
L_{wire1}	0.0889 nH
R_s	4.00 Ω
C_s	8.19 pF

TABLE II
EXTRACTED PARAMETERS WITH DIFFERENT DC BIAS CURRENTS OF THE EQUIVALENT CIRCUIT FOR INTRINSIC DFB LASER SHOWN IN FIG. 3

Injection Current (mA)	C_t(pF)	R_1(Ω)	R_{s1}(mΩ)	R_{s2}(μΩ)	L_s(pH)
15.9	707	0.721	17.6	17.9	1.43
20.9	450	0.721	17.6	17.9	1.24
23.9	392	0.721	17.6	17.9	1.15
36.9	413	0.721	17.6	17.9	0.641

laser chip can be represented as a series resistance and a contact capacitance, which are modeled by R_s and C_s, respectively. The series resistance consists of a contact resistance and a substrate resistance, and the contact capacitance is originated from reverse-biased current blocking junction and ohmic metal-insulator-semiconductor layer. The inductance and the capacitance, $L_{\rm wire1}$ and $C_{\rm cpw1}$, due to bonding wires and a tapered coplanar waveguide are included in the circuit model.

In this circuit model, the small signal photon density is proportional to the output voltage across R_{s2}. Hence the output voltage across R_{s2} can be used as a measure of optical intensity and an output port for measuring S parameters. Using measured several frequency responses, S_{21}, with different dc bias currents of DFB laser on submount, each parameter of the circuit model was extracted. The extracted parameters of the equivalent circuit elements are shown in Tables I and II. Each parameter in Table II should be different because of different frequency responses depending on dc bias currents except for resistance values. The resistance values do not change with varying dc bias currents. Fig. 4 shows that the small signal frequency responses calculated from the equivalent circuit (Fig. 3) with the extracted parameters agree well with those measured from the DFB laser on submount. The peak of each frequency response is located at higher frequency with increasing dc bias currents. For calculating frequency responses depending on dc bias currents, we varied the only parameters related to the rate equation. We fixed the parameters related to the parasitic components due to the semiconductor laser chip, bonding wires and tapered coplanar waveguide. The different relaxation oscillations and roll-off effects of frequency responses due to the different biasing conditions were well predicted from the equivalent circuit model using extracted parameters from measured S parameters.

D. Equivalent Circuit of Complete Cooled Butterfly 2.5 Gbps DFB Laser Modules

For a complete equivalent circuit of 2.5 Gbps DFB laser modules, we combined the equivalent circuit model of DFB laser on submount with parasitic components due to butterfly package, TEC, coplanar waveguide, and bonding wires. Fig. 5 shows a complete equivalent circuit of 2.5 Gbps DFB laser modules. For the coplanar waveguide with a thin-film resistor can be modeled by the two capacitances, $C_{\rm cpw2}$ and $C_{\rm cpw3}$, because of its short length. The thin-film resistor used to match 25 Ω impedance was represented to inductance L_{rs}, and resistance R_r (20 Ω) [7]. The end of this coplanar waveguide is connected to the tapered coplanar waveguide on DFB laser on submount through bonding wires, $L_{\rm wire4}$ and $L_{\rm wire5}$. The ground of the coplanar waveguide in butterfly-type package is connected to the case ground of butterfly-type package through a chip capacitor of 100 pF, C_g, and bonding wires, $L_{\rm wire6}$ and $L_{\rm wire7}$. Parasitic components of TEC proposed by Hiroyuki were used [8]. Inductances L_i and L_o are due to the leads of

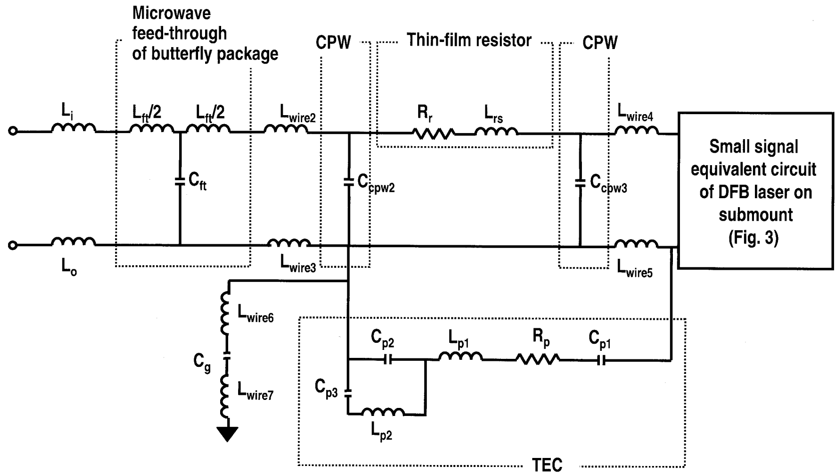


Fig. 5. Small-signal equivalent circuit model of cooled butterfly-type 2.5 Gbps DFB laser module.

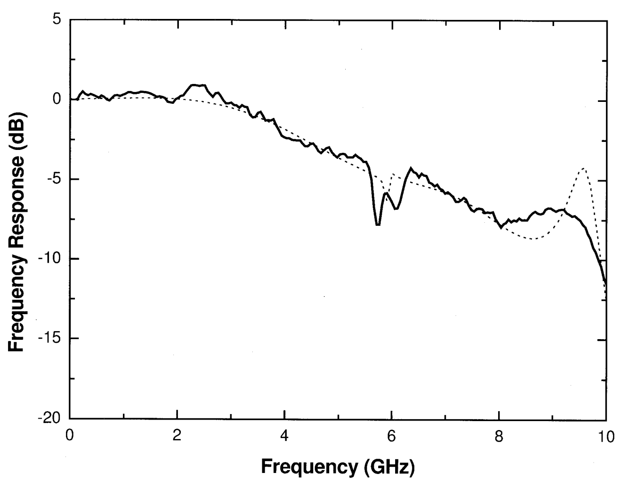


Fig. 6. Frequency responses of 2.5 Gbps DFB laser modules. Solid line represents measured data and dotted line represents simulated data using the small-signal equivalent circuit model.

TABLE III
EXTRACTED PARAMETERS OF THE EQUIVALENT CIRCUIT FOR COOLED BUTTERFLY-TYPE 2.5 Gbps DFB LASER MODULES SHOWN IN FIG. 5

Parameter	Description	Value/Unit
L_i	External lead inductance	0.570nH
L_o	External lead inductance	0.570nH
L_{ft}	Inductance from feed-through	0.309 nH
C_{ft}	Capacitance from feed-through	1.13 pF
L_{wire2}	Inductance from bonding wire	0.116 nH
L_{wire3}	Inductance from bonding wire	0.113 nH
C_{cpw2}	Capacitance from CPW	2.76pF
R_r	Resistance from thin film resistor	20.0 Ω
L_{rs}	Inductance from thin film resistor	0.0470nH
C_{cpw3}	Capacitance from CPW	0.719pF
L_{wire4}	Inductance from bonding wire	0.417nH
L_{wire5}	Inductance from bonding wire	0.0876nH
L_{wire6}	Inductance from bonding wire	0.120 nH
L_{wire7}	Inductance from bonding wire	0.0985 nH
C_g	Capacitance from chip capacitor	100pF
C_{p1}	Capacitance from TEC	2.00 pF
C_{p2}	Capacitance from TEC	2.00 pF
C_{p3}	Capacitance from TEC	0.250 pF
L_{p1}	Inductance from TEC	0.100 nH
L_{p2}	Inductance from TEC	3.00 nH
R_p	Resistance from TEC	2.00 Ω

microwave signal feed-line in butterfly-type packages. The microwave feed-through of butterfly-type packages can be modeled with $L_{\rm ft}$ and $C_{\rm ft}$. The coplanar waveguide in butterfly-type packages is connected to a microwave feed-through of butterfly packages through a couple of bonding wires to reduce inductance. These bonding wires are included by $L_{\rm wire2}$ and $L_{\rm wire3}$.

Using the measured several frequency response, S_{21}, of the complete DFB laser modules with the bias current of 23.9 mA, each parameter of the circuit model was extracted except for DFB laser on submount. The small-signal circuit parameters of DFB laser on submount with the bias current of 23.9 mA were used in order to extract parameters of modules. The extracted parameters of the equivalent circuit elements are listed in Table III. Fig. 6 shows the measured frequency response of the DFB laser module and the calculated frequency response of the equivalent circuit model in Fig. 5. These results show that the calculated frequency response using the extracted parameters is similar to the measured frequency response. These frequency responses of DFB laser modules show that high-speed characteristics of the DFB laser module can be highly dependent on parasitic components due to packages rather than the semiconductor laser chip and its parasitic components. Especially, the 3dB frequency bandwidth restricted by parasitic components of DFB laser modules. In the measured frequency responses, a dip was observed at around 6 GHz. This dip was also observed in the calculated frequency response using the equivalent circuit model. Without including parasitic components of TEC, the dip did not appear in the frequency response. So, this dip is due to parasitic components of TEC.

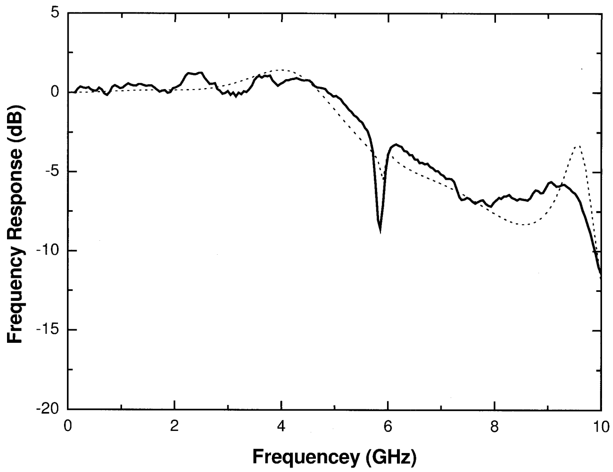


Fig. 7. Improved frequency responses of 2.5 Gbps DFB laser modules using the proposed method. Solid line and dotted line represent measured data and simulated data, respectively.

III. Improvements of Frequency Responses for 2.5 Gbps DFB Laser Modules Using the Equivalent Circuit Model

To achieve high performance operation of DFB laser modules at high speed, the 3 dB frequency bandwidth of DFB laser modules should be extended up to 5 GHz. It is required to extend the 3 dB frequency bandwidth using optimizing frequency responses of DFB laser modules. It is obvious that reducing parasitic components of packages can improve frequency responses. This method is not easy to implement. Using the equivalent small-signal circuit model of 2.5 Gbps DFB laser modules, we found that the capacitance of the coplanar waveguide between matching resistor and microwave feed-through in butterfly package has the largest influence on the frequency response, compared with other components. But, reducing the capacitance of the coplanar waveguide is not useful, because this method can add other parasitic components. Another method to improve the frequency response is to add a chip capacitor across the thin-film resistor. We attached a 1 pF chip capacitor across the 20 Ω thin-film resistor. Because this capacitor provides a signal path at high frequency, the high frequency response of DFB laser module can be improved. To model this effect, we used the chip capacitor model [7] composed of parasitic resistance, conductance, inductance and capacitance. Fig. 7 shows the measured and calculated S_{21} after adding the chip capacitor across the thin-film resistor. This high frequency signal path through the chip capacitor improved the 3 dB frequency bandwidth from 3 GHz to 5 GHz. This chip capacitor does not affect the frequency response below 2.5 GHz, because at the low frequency, the capacitance is operated as open circuit. But, this chip capacitor provides signal path to electrical signals over 3 GHz. This method is very simple and useful to improve the frequency characteristics by adding a chip capacitor. Using the equivalent circuit model, we can predict frequency responses of DFB laser modules during manufacturing process. Therefore, it can be used to predict and improve frequency characteristics whenever the package design is made.

IV. Conclusion

We successfully demonstrated the equivalent small-signal circuit model of 2.5 Gbps cooled butterfly-type DFB laser modules. The modified two-port black box model was used to model the equivalent circuit of 2.5 Gbps DFB laser modules. At first, DFB laser on submount was modeled using measured S parameters. And then, the complete 2.5 Gbps DFB laser module was modeled based on the model of DFB laser on submount. Using the complete equivalent small-signal circuit model, we proposed and demonstrated the method to improve frequency characteristics of 2.5 Gbps DFB laser modules through experiments and simulations.

References

[1] I. Tomkos, D. Chowdhury, J. Conradi, D. Culverhous, K. Ennser, C. Giroux, B. Hallock, T. Kennedy, A. Kruse, S. Kumar, N. Lascar, I. Roudas, M. Sharma, R. S. Vodhanel, and C.-C. Wang, “Demonstration of negative dispersion fibers for DWDM metropolitan area networks,” *IEEE J. Selected Topics Quantum Electron.*, vol. 7, pp. 439–460, May/June 2001.

[2] F. Delpiano, R. Paoletti, P. Audagnotto, and M. Puleo, “High frequency modeling and characterization of high performance DFB laser modules,” *IEEE Trans. Comp., Manufact., Packag., Technol. B*, vol. 17, pp. 412–417, Aug. 1994.

[3] S.-S. Park, M. K. Song, S. G. Kang, N. Hwang, H. T. Lee, H. R. Choo, and K. E. Pyun, “High frequency modeling for 10 Gbps DFB laser diode module packaging,” in *Proc. 46th Electron. Comp. Technol. Conf.*, May 28–31, 1996, pp. 884–887.

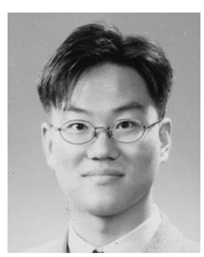
[4] R. S. Tucker, “High-speed modulation of semiconductor lasers,” *IEEE J. Lightwave Technol.*, vol. LT-3, pp. 1180–1192, Dec. 1985.

[5] R. S. Tucker and I. P. Kaminow, “Height-frequency characteristics of directly modulated InGaAsP ridge waveguide and buried heterostructure lasers,” *IEEE J. Lightwave Technol.*, vol. LT-2, pp. 385–393, Aug. 1984.

[6] R. S. Tucker and D. J. Pope, “Circuit modeling of the effect of diffusion on dampling in a narrow-stripe semiconductor laser,” *IEEE J. Quantum Electron.*, vol. QE-19, pp. 1179–1183, July 1983.

[7] M. Golio, *The RF and Microwave Handbook*. Orlando, FL: CRC Press, 2001.

[8] H. Nakano, S. Sasaki, M. Maeda, and K. Aiki, “Dual-in-line laser diode module for fiber-optic transmission up to 4Gbit/s,” *IEEE J. Lightwave Technol.*, vol. LT-5, pp. 1403–1411, Oct. 1987.



Jaehoon Lee received the B.S. degree in radio engineering from Korea University, Seoul, in 1997, the M.S. degree in electrical engineering from the Korea Advanced Institute of Science and Technology, Taejon, in 1999, and is currently pursuing the Ph.D. degree in radio engineering at Korea University.

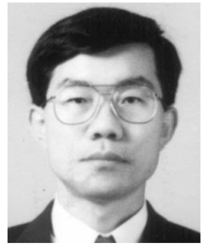
His current research interests include the packaging of the module for high-speed optical communication, modeling of fiber optic communication systems, and modulation formats for DWDM system.

Seungki Nam received the B.S. and M.S. degrees in radio engineering from Korea University, Seoul, in 1998 and 2000, respectively, where he is currently pursuing the Ph.D. degree in radio engineering.

His research interests are electrical characteristics of high-speed electric and optoelectronic device packages and modeling of high-speed interconnections.

S. H. Lee received the B.S. degree in electronics engineering from Chung-Ang University, Seoul, Korea, in 1989 and the M.S. degree in radio sciences and engineering from Korea University, Seoul, in 2000.

He worked for seven years at Samsung Advanced Institute of Technology, Suwon, Korea, where he was Member of Technical Staff, Optoelectronic Laboratory. In 1999, he joined Samsung Electronics' Optoelectronics Division, Suwon, and is now a Chief Engineer of the Optic Packaging Group. His research interests include RF matching circuits at high-speed optic package and thermal behavior in optic devices.



Jichai Jeong (SM'96) received the B.S. degree from Korea University, Seoul, in 1980, the M.S. degree from the Korea Advanced Institute of Science and Technology (KAIST), Taejon, in 1982, and the Ph.D. degree in electrical engineering from Carnegie Mellon University, Pittsburgh, PA, in 1988.

From 1982 to 1985, he was a Researcher at KAIST. From 1988 to 1993, he was Member of Technical Staff, AT&T Bell Laboratories, Murray Hill, NJ, where he worked in optoelectronic integrated circuits and semiconductor lasers for optical communications. From 1993 to 1995, he was the faculty of the Electrical Engineering Department, Pohang University of Science and Technology. In 1995, he joined the faculty of the Radio Engineering Department, Korea University. His current research interests include modeling and simulation of optical components and transmission systems for fiber optic communications.